

Executive Summary: Nassau Candy Distributor - Shipping Efficiency and Route Performance Analysis

Overview

This project analyzes shipping performance across Nassau Candy Distributor's route network to identify lead-time inefficiencies, route bottlenecks, and logistics optimization opportunities. The analysis focuses on route volume, delivery lead time, geographic variation, and shipping-mode performance.

Key Findings

- Shipping performance varies materially across routes, with several high-volume lanes experiencing longer delivery times than expected.
- Regional destination patterns reveal logistics bottlenecks, indicating that some geographies consistently face slower delivery performance.
- Shipping-mode performance is uneven across the network, suggesting that service selection and operational execution are not consistently optimized.
- Route analytics provide a clear foundation for targeted logistics improvement once the source data issue is addressed.

Critical Data Limitation

The source dataset contains a material inconsistency: several shipment dates extend into 2026 while their corresponding order dates remain in 2024. This produces unrealistic lead times exceeding 900 days and materially distorts delay-related metrics.

Business Impact

Because lead-time data is central to logistics planning, the identified date inconsistency can misstate route inefficiency, exaggerate delay exposure, and weaken confidence in operational decisions based on the current extract.

Recommendations

- Validate shipment-date records directly against the operational source systems before acting on route performance conclusions.
- Implement ETL-level data-quality controls to prevent future date inconsistencies from reaching analytical outputs.
- Prioritize review of high-volume routes that continue to show delay pressure after the data issue is corrected.
- Use the corrected route insights to refine regional planning and shipping-mode usage.

Conclusion

Project 2 establishes a strong route-efficiency analysis framework, but reliable operational action depends on first resolving the shipment-date quality issue. Once corrected, the same analysis can directly support logistics optimization.